EV and the Grid HW Feb13
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ME4803/8803 EV Charging Homework
2/18/2026

1) Optimized Charging for your "City/State". All of the questions in this problem refer to the city/state that you have picked earlier in the semester.

a) What is the City/State that you have picked?

Cedar City, Utah, USA

b) What is the cost of electricity (per kWh) in that city.

Average residential electricity cost in Cedar City **~13.07 ¢/kWh** (based on recent local utility data) [1].

Statewide average electricity cost for Utah is around **~14 ¢/kWh** as of early 2026 [2].

c) What are the Time of Use (TOU) rates? Usually these are done at the electric utility level, so you may have to choose a specific electric utility in the city (likely the largest utility). What are the details/times and rates for the TOU offering?

- On-peak: **~32.0834 ¢/kWh** (June–Sep) / **~28.3924 ¢/kWh** (Oct–May)
- Off-peak: **~7.1296 ¢/kWh** (June–Sep) / **~6.3094 ¢/kWh** (Oct–May)
- Weekday on-peak hours: **6 p.m. – 10 p.m.**
  (*Plan effective Dec 1 2025*) [3].

Additional TOU Options are available with wider ranges (e.g., **7¢ – 16¢ / kWh**, or **3¢ – 33¢ / kWh** for EV TOU pilot plans) depending on schedule chosen [4].

d) How does the utility generate its electricity (from what sources)? Can you locate the percentages and values for the region?

Estimated electricity generation mix in 2023-2024 is approximately
(Net generation percentages from 2023 data) [5].

i) Coal __47%__
ii) Nat Gas __35%__
iii) Solar __11.6%__
iv) Wind __2%__
v) Hydro __1.6%__
vi) Nuclear __0%__
vii) Other(geothermal/biomass) __2%__

e) Locate any information on how the carbon footprint for electricity generation changes by the hour in the region? If you can't locate specifics from the utility or ISO, what conclusions can you infer from question (d) above on how CO2 might vary by hour? Explain.

- Utah's grid mix is **heavily dependent on fossil fuels** (coal & gas). These sources produce more CO_2 per kWh than renewables.
- **Renewable output (solar) typically peaks daytime**, reducing CO_2 intensity in mid-day hours.
- **Evening/late-day demand relies more on coal and natural gas**, raising CO_2 intensity (because renewables generally decline when solar output drops).
- Without precise hourly data, one can infer **higher CO_2 intensity** in late afternoon/evening (**peak demand**) and **lower intensity midday** when **solar** contributes more generation.

f) Locate any demand or generation graphs by hour for the region's electric grid. Explain / Interpret it.

Current hourly generation graphs for Utah aren't directly provided, nor could I find any graph for the hourly pattern for this city, but statewide trends are well documented so we will try to reason what it could most likely look like.

Typical daily grid behavior in mixed fossil/solar regions:

- Solar energy production peaks around **mid-day**, reducing fossil generation immediately then.
- Afternoon/evening peaks occur as solar declines before day ends, requiring more dispatchable fossil generation (natural gas/coal).
- This results in a *"duck curve"-like* pattern widely observed in grids with significant solar capacity [6].

g) Explain unique factors that you located relative to electricity generation for the utility? For example, is severe weather an issue? How much of the energy is used for cooling or heating homes? What is the percentage of residential electric demand vs industrial vs commercial?

Weather Impact:

Utah experiences **hot summers and cold winters**, significantly driving electricity demand for both air conditioning and heating—especially peak load during summer afternoons and early evenings. Peaks have historically **strained grid infrastructure due to large temperature swings** [7].

Residential vs Commercial vs Industrial Demand:

Based on state energy profiles (see Pew research), in 2022 Utah's electricity use by sector was[8]:
Commercial: ~38.6%
Residential: ~34%
Industrial: ~27.3%
Transportation uses are very minimal (<1%) [8].

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2) Using the information discovered above, what is an **optimal charging algorithm for an EV located in that city/region?** Feel free to get creative. For example, if your city has a high percentage of wind, you could write an algorithm that optimizes the charging based on wind speeds by the hour. You could optimize the algorithm for economics (i.e. price/cost), CO2, weather, electric load, etc. Write your algorithm approach below, and rationale. Provide calculations and in-depth explanation in your response.

Algorithm Approach

The algorithm is a **greedy, time-slot–based** optimizer over a single overnight charging window. It uses **15-minute** intervals and only considers slots between user-defined plug-in and departure times. Three optimization modes are supported [9], [10]:

1. **Cost** – Minimize electricity cost using time-of-use (TOU) prices.
2. **CO2** – Minimize carbon by preferring hours with lower CO2 intensity (lb/kWh).
3. **Mixed** – Minimize a weighted score: 70% normalized price + 30% normalized CO2.

Rationale

- **TOU:** Peak periods (e.g., 6–10 PM) are typically more expensive and often more carbon-intensive; shifting charging to off-peak reduces cost and can reduce emissions [9], [11].
- **Seasonal profiles:** Demand and generation mix vary by season (e.g., more solar in summer daytime, higher heating demand at night in winter). The implementation uses season-specific TOU and CO2 profiles (Winter, Spring, Summer, Fall) [12].
- **Mixed mode:** Balances economic and environmental goals when the cheapest hours are not the cleanest [10].

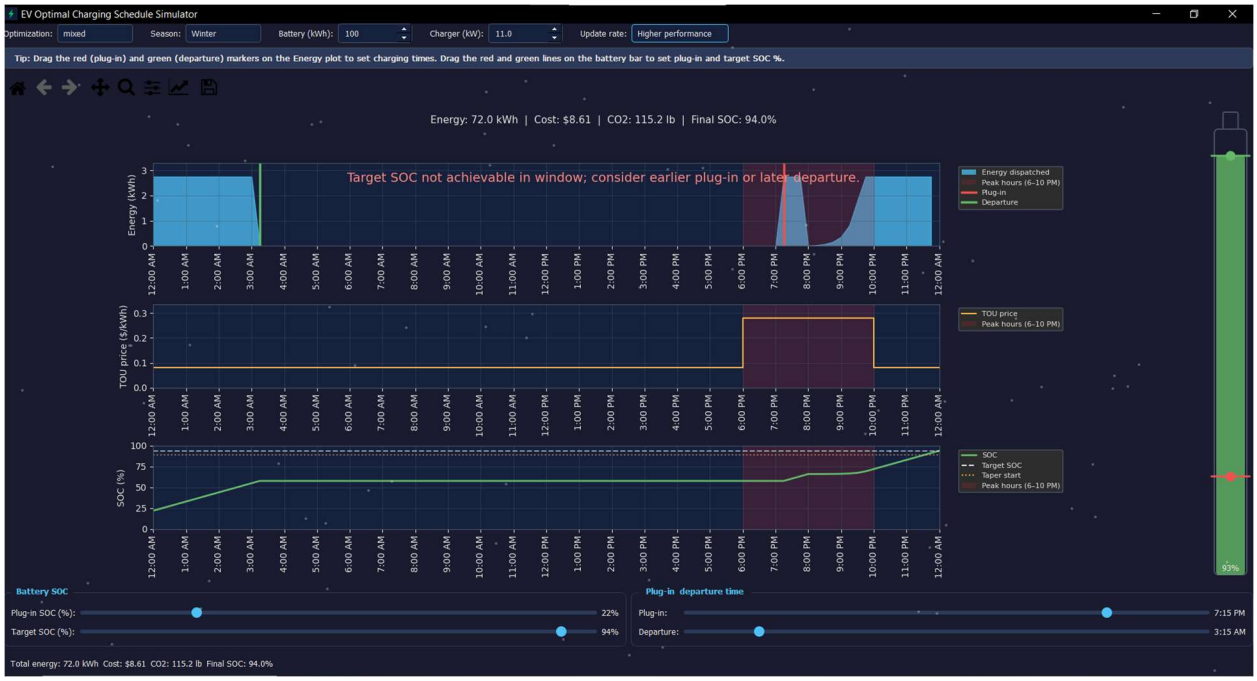
Calculations

- **Energy needed (kWh)** E_need = battery capacity (kWh) × (target SOC − plug-in SOC) / 100.

- **Power taper (Tesla-like)** Above a “taper start” (target − 5%), power ramps down linearly to zero at target SOC to model real-world charging behavior [13]: P = P_max × (target − SOC) / (target − taper start).
- **Greedy rule** Among slots in the plug-to-departure window, sort by price (cost mode), CO2 (CO2 mode), or weighted score (mixed). Charge in that order, filling each 15-min slot up to the taper-limited power and energy needed until the target SOC is met.
- **Totals** Cost: Σ (kWh per slot) × (price per slot). CO2: Σ (kWh per slot) × (CO2 intensity per slot).

In-Depth Explanation

The region is modeled with on-peak (e.g., 6–10 PM) and off-peak rates, and with day (e.g., 6 AM–6 PM) vs night CO2 intensities that vary by season [12]. The algorithm does not change how much energy is requested (that is fixed by the user's target SOC) but only when within the plug-in window that energy is taken. By always choosing the “best” remaining 15-min slot (cheapest, cleanest, or best mixed score), it minimizes cost or CO2 (or a mix) for that session, subject to the battery's power taper [13]. This fits regions with strong TOU and time-varying grid mix (solar, wind, heating demand) and can be extended with real wind/solar forecasts by using hourly CO2 or price signals derived from those forecasts [14].



Note: Originally, I attempted to develop the algorithm in MATLAB, but I encountered several issues. As a result, I pivoted and created a Python application for Windows that runs as a standalone executable file.

The application allows you to modify various settings, including the season, plug-in time, initial state of charge, desired state of charge, departure time, battery capacity, charger rate, and objective optimization. However, the electricity rates are currently defaulted to Cedar City, Utah.

I have included the GitHub link here because the project file exceeds 2K lines. The main source file is **ev_charging_simulator.py**, and the compiled Windows application (.exe) can be found in the **dist** folder.

The repository is public in case anyone would like to expand it by adding additional factors or improvements.

GitHub Repo: https://github.com/hpark416/EVchargeSimulation/tree/main/dist

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3) CO2 Calculation (and cost) for driving electric in your city/state.

a. Using **eia.gov**, determine the cost per kWh and the CO2 per kWh for electricity.

____~$0.10_____ $/kWh _____~1.54____ lbs. CO2/kWh

Average residential price: ~$0.0997/kWh (all sectors weighted) [15].

According to Climatiq/eGRID data, electricity supplied from the Utah grid emits ~0.691–0.710 kg CO_2 per kWh [16]

Converting kilograms to pounds: 1 kg CO_2/kWh $\approx$ 2.20462 lbs CO_2/kWh

Therefore:

0.69 kg/kWh × 2.20462 ≈ 1.52 lbs CO_2/kWh

0.71 kg/kWh × 2.20462 ≈ 1.57 lbs CO_2/kWh

We'll use the mid-range value for a reasonable average: ~1.54 lbs CO_2/kWh

Utah's generation mix is still dominated by fossil fuels (coal ~47%, natural gas ~35%), with relatively small contributions from renewables.[8]

b. Compare the two vehicles below in terms of cost and CO2 for 100 miles of driving. Use the average price of gasoline in the state. List your sources and values. Explain the differences and why you think they have the values listed.

i. *2024 Toyota Corolla 2.0L Gasoline*

_____8.13_____ $/100 Miles _____56.0_____ lbs. CO2/100 Miles

ii. *2025 Hyundai Ioniq 6 EV (Standard Range)*

_____2.49_____ $/100 Miles _____30.7_____ lbs. CO2/100 Miles

Gasoline price (UT avg): **$2.846/gal** [17]

Corolla fuel economy: **35 mpg combined** [18]

Gasoline CO_2 factor: **19.6 lb CO_2/gal** [19]

Utah electricity price: **$0.0997/kWh** [20]

Utah electricity CO_2 intensity: **~1.23 lb CO_2/kWh** [21]

Ioniq 6 efficiency: **135 MPGe combined** [22]

(1 gal gasoline = 33.7 kWh [19])

2024 Toyota Corolla (Gas)

Fuel used: $100/35 = 2.857$ gal

Cost: $2.857 \times 2.846 = \$\mathbf{8.13}$

CO_2: $2.857 \times 19.6 = \mathbf{56.0}$ lb CO_2

2025 Hyundai Ioniq 6 (EV)

Energy use: $100 \times (33.7/135) = 24.96$ kWh

Cost: $24.96 \times 0.0997 = \$\mathbf{2.49}$

CO_2: $24.96 \times 1.23 = \mathbf{30.7}$ lb CO_2

The **EV costs ~69% less per 100 miles** because Utah electricity (~$0.10/kWh) is much cheaper per unit of usable driving energy than gasoline at $2.846/gal.

The **EV emits ~45% less CO_2 per 100 miles**. Although Utah's grid still relies heavily on fossil fuels (coal + natural gas), generating electricity is still more energy-efficient overall than burning gasoline in an internal combustion engine. Gas vehicles emit CO_2 directly from combustion, while EV emissions depend on grid mix. As Utah adds more renewables, EV emissions will decrease further.

c. Explain how the EV cost or CO2 compares to the national averages (how does driving in the state compare to the national average for greenhouse gas and cost)

- **Utah average electricity price:** ~$0.0997/kWh [15]
- **U.S. national average electricity price:** ~$0.157/kWh [23]

Because Utah's electricity rate is lower than the national average, EV charging in Utah costs less per mile.
For the Ioniq 6 (≈24.96 kWh per 100 miles):

- **Utah cost:**
  24.96 × 0.0997 ≈ **$2.50 per 100 miles**
- **U.S. average cost:**
  24.96 × 0.157 ≈ **$3.92 per 100 miles**

EV driving in Utah is cheaper than the U.S. average due to lower electricity rates.

CO_2 Emissions Comparison

- **Utah grid CO_2 intensity:** ~1.23 lb CO_2/kWh [16]
- **U.S. average grid CO_2 intensity:** ~0.85 lb CO_2/kWh [24]

For 100 miles:

- **Utah EV emissions:**
  24.96 × 1.23 ≈ **30.7 lb CO_2**
- **U.S. average EV emissions:**
  24.96 × 0.85 ≈ **21.2 lb CO_2**

Utah's EV emissions are higher than the national EV average because the state relies more heavily on coal and natural gas for electricity generation [8].

- **Cost:** Utah EV driving is **less expensive** than the national average.
- **Emissions:** Utah EV driving produces **more CO_2 than the national EV average**, but still less than gasoline vehicles.

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4) Charge Couplers

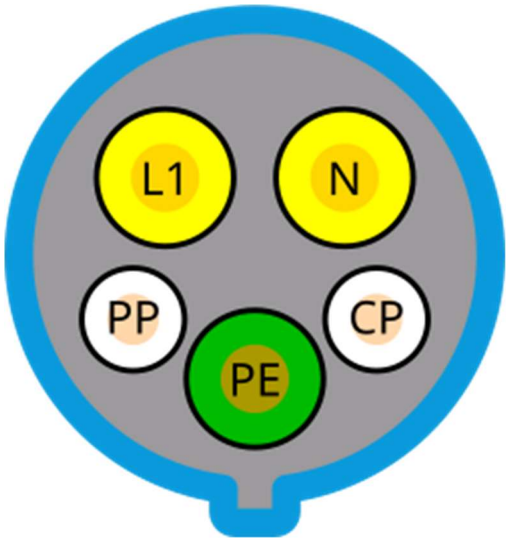
4a) Identify each of the three couplers below:

[AC] J1772(Type1) Coupler | *[AC/DC] Tesla NACS Coupler* | *[AC] GB/T*

Starting from left to right, the couplers are identified by their shape, number of pins, and whether the charging socket uses male or female contacts, as shown above. When matched with the reference image, the first two connectors are easily identified. The third connector was slightly more difficult to distinguish because the MENNEKES and GB/T standards appear similar. The key difference lies in the nature of the pin contacts in the charging ports—MENNEKES connectors use male contacts, whereas the GB/T standard uses female contacts. Therefore, the last charging socket is a GB/T connector due to the presence of female contacts rather than male contacts typically found in a MENNEKES connector [25].

4b) Label the pins for the Couplers Below:

These plugs are a J1772 Type1 [26] and a Tesla plug [27] reference images and corresponding pinout table are shown below.



L1	Line 1	single-phase AC
L2/N	Line 2 / Neutral	single-phase AC
CP	Control pilot	post-insertion signaling
PP	Proximity pilot	pre-insertion signaling
PE	Protective earth	full-current protective earthing system

DC+/L1	DC+ / Line 1	Positive current (DC), Line 1 (split phase AC), or Line (single phase AC)
DC−/L2	DC− / Line 2	Negative current (DC), Line 2 (split phase AC), or Neutral (single phase AC)
G	Ground	Protective grounding system

CP	Control pilot	Charging state/current signaling
PP	Proximity pilot	Vehicle connector status signaling

4c) Fill in the table for the voltage for each state below for **J1772**

The data was sourced from SAE J1772 Pilot Signal State Voltages, Electric Vehicle Charging Equipment, Utah Dept. of Commerce [28].

State	Pilot Signal (Max Voltage)	Meaning (self-added)
EV Disconnected	12V	No EV connected (State A)
EV Connected	~9V	EV plugged in but not charging (State B)
EV Charge	~6V	EV is ready and actively charging (State C)

These voltage levels are derived from typical SAE J1772 pilot signal interpretations where the EVSE drives a 1 kHz PWM ±12 V square wave on the control pilot and the vehicle loads it down with resistances to signal states.

- The "max voltage" shown here is the approximate **pilot high level** measurement for each state as defined in SAE J1772 signaling tables.
- Beyond these three, additional states include ventilation required (~3 V), fault (0 V), and error states (–12 V).

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